

### i When to Use

To compare the means of two or more quantitative variables obtained from dependent samples (repeated measures or matched groups). The two or more scores might be the same variable measured at different times or under different conditions, comparable variables measured at the same time, or some combination.

**H Research Hypothesis:** There are significant differences in the three types of engagement scores (emotional impact, aesthetic appeal, cognitive engagement) for Superman media portrayals, with emotional impact receiving the highest scores and cognitive engagement receiving the lowest.

**H Null Hypothesis:** Superman media portrayals have the same mean scores for emotional impact, aesthetic appeal, and cognitive engagement.

## Research Design

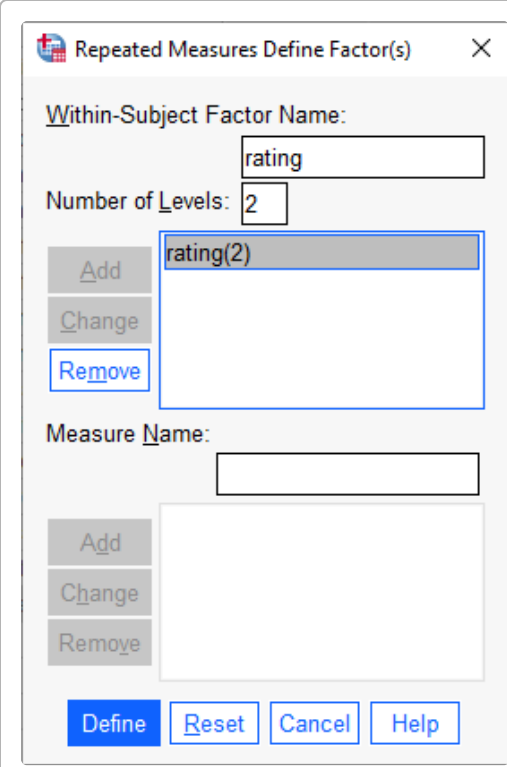
The IV is Engagement Type, with the conditions Emotional Impact, Aesthetic Appeal & Cognitive Engagement. The DV is the score on each engagement measure.

| Emotional Impact | Aesthetic Appeal | Cognitive Engagement |
|------------------|------------------|----------------------|
|                  |                  |                      |

**Variables in the Analysis:** In a WG design the variables in the analysis are the variables holding the DV scores for each IV condition (emotional\_impact, aesthetic\_appeal & cognitive\_engagement).

**Analyze → General Linear Model → Repeated Measures**

- In the **Repeated Measures Definition Window** enter your name for the IV in the “Within-subject Factor Name” box (EngagementType)
- Enter the number of conditions of the IV in the “Number of levels” window (3)
- Click the “Add” button
- Click the “Define” button
- In the **Repeated Measures** window highlight the variables that are the DV score for each condition and click the arrow
- Click the “Options” button — in the **Repeated Measures: Options** window check the “Descriptives” box



**Repeated Measures Define Factor(s)**

Within-Subject Factor Name:

Number of Levels:

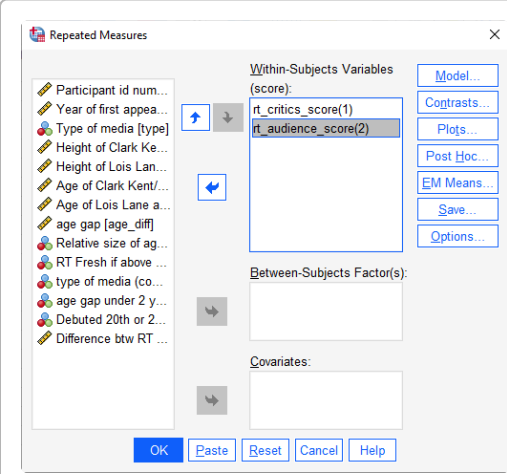
Add Change Remove

rating(2)

Measure Name:

Add Change Remove

Define Reset Cancel Help



**Repeated Measures**

Participant id num...  
Year of first appea...  
Type of media [type]  
Height of Clark Ke...  
Height of Lois Lan...  
Age of Clark Kent/...  
Age of Lois Lane a...  
age gap [age\_diff]  
Relative size of ag...  
RT Fresh if above ...  
type of media (co...  
age gap under 2 y...  
Debuted 20th or 2...  
Difference btw RT ...

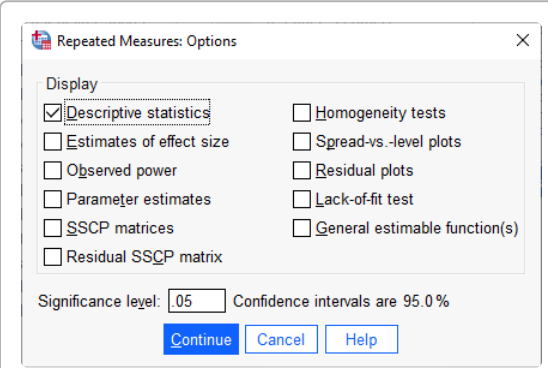
Within-Subjects Variables (score):  
rt\_critics\_score(1)  
rt\_audience\_score(2)

Model...  
Contrasts...  
Plots...  
Post Hoc...  
EM Means...  
Save...  
Options...

Between-Subjects Factor(s):

Covariates:

OK Paste Reset Cancel Help



**Repeated Measures: Options**

Display

☒ Descriptive statistics ☐ Homogeneity tests

☐ Estimates of effect size ☐ Spread-vs.-level plots

☐ Observed power ☐ Residual plots

☐ Parameter estimates ☐ Lack-of-fit test

☐ SSCP matrices ☐ General estimable function(s)

☐ Residual SSCP matrix

Significance level:  Confidence intervals are 95.0 %

Continue Cancel Help

SPSS Syntax

```
GLM emotional_impact aesthetic_appeal cognitive_engagement /WSFACTOR=EngagementType 3
/PRINT=DESCRIPTIVE.
```

- ① List variables holding DV for each IV condition
- ② Give a name to the WG IV & tell # conditions
- ③ Get mean and std

## SPSS Output

### Descriptive Statistics

|                      | Mean  | Std. Deviation | N  |
|----------------------|-------|----------------|----|
| emotional_impact     | 13.27 | 3.385          | 44 |
| aesthetic_appeal     | 9.80  | 2.833          | 44 |
| cognitive_engagement | 4.36  | 1.170          | 44 |

### Tests of Within-Subjects Effects

Measure: MEASURE\_1

| Source                   | Type III Sum of Squares         | df      | Mean Square | F       |
|--------------------------|---------------------------------|---------|-------------|---------|
| <b>Sig.</b>              |                                 |         |             |         |
| EngagementTypeSphericity | 1776.110                        | 2       | 888.055     | 138.518 |
| Assumed                  |                                 |         |             |         |
| <.001                    | Error(EngagementTypeSphericity) | 551.357 | 86          | 6.411   |
|                          | Assumed                         |         |             |         |

SPSS provides different “versions” of the ANOVA output. We will focus on the “traditional” analysis, which SPSS labels as “Sphericity Assumed”.

- **df effect & F & p-value** to use
- **df(error) & Mean Square Error** term for this analysis

There is a significant difference among the means of the three different types of engagement scores,  $F(2, 86) = 138.518, p < .001$ .

## Steps for computing and interpreting LSD minimum mean difference

1. Determine df(error) for the analysis: **df(error) = 86**
2. Use the t-table to determine the critical value of t (at  $p = .05$ ) for this df(error): **t-critical = 1.99**
3. Determine the MSerror for the analysis: **MSerror = 6.41**
4. Determine the number of participants in the analysis: **n = 44**
5. Apply the LSD formula to obtain the minimum mean difference

$$d_{LSD} = \frac{t \times \sqrt{2 \times MSE_{error}}}{\sqrt{n}} = \frac{1.99 \times \sqrt{2 \times 6.41}}{\sqrt{44}} = 1.07$$

6. Compare the minimum mean difference (1.07) with each pairwise mean difference:
  - If the pairwise mean difference is larger than the minimum mean difference, then those two conditions have means that are “significantly different”
  - If the pairwise mean difference is smaller than the minimum mean difference, then those two conditions have means that are “statistically equivalent”

**For this analysis** — compare the minimum mean difference of 1.07 with each of the pairwise differences:

| Pair                                     | Mean Difference | LSD Result  |
|------------------------------------------|-----------------|-------------|
| Emotional Impact vs Aesthetic Appeal     | 3.48            | Significant |
| Emotional Impact vs Cognitive Engagement | 8.91            | Significant |
| Aesthetic Appeal vs Cognitive Engagement | 5.44            | Significant |

## Reporting the Results

### Sample APA Write-Up

The engagement scores for Superman media portrayals are summarized in Table 1. Emotional Impact ( $M = 13.27$ ,  $SD = 3.39$ ), Aesthetic Appeal ( $M = 9.80$ ,  $SD = 2.83$ ), Cognitive Engagement ( $M = 4.36$ ,  $SD = 1.17$ ). There was a significant difference among the three engagement score distributions,  $F(2,86) = 138.52$ ,  $MSe = 6.41$ ,  $p < .001$ . Pairwise comparisons using LSD (with a minimum mean difference = 1.07) revealed that Emotional Impact was rated higher than Aesthetic Appeal, and Emotional Impact was rated higher than Cognitive Engagement, and Aesthetic Appeal was rated higher than Cognitive Engagement.

**Table 1.**

Summary of engagement scores for Superman media portrayals.

|                      | Mean  | SD   |
|----------------------|-------|------|
| Emotional Impact     | 13.27 | 3.39 |
| Aesthetic Appeal     | 9.80  | 2.83 |
| Cognitive Engagement | 4.36  | 1.17 |